

WAKAYAMA, Japan

WMO: 477770

Lat: 34.2289N

Lon: 135.1642E

Elev: 18

StdP: 101.10

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.4	1.3	-8.0	1.9	4.3	-6.8	2.1	4.8	12.2	5.9	11.0	6.2	4.0	70	0.450

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.7	33.6	24.9	32.5	24.9	31.5	24.8	26.6	31.0	26.1	30.5	25.6	30.0	4.7	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.3	20.5	29.3	24.8	19.8	29.0	24.3	19.2	28.7	83.1	31.1	81.1	30.6	79.1	30.1	28.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.9	9.2	8.0	-1.3	35.8	0.7	1.1	-1.8	36.6	-2.2	37.2	-2.6	37.9	-3.1	38.7
			-3.3	27.1	0.9	0.8	-3.9	27.7	-4.5	28.1	-5.0	28.6	-5.7	29.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.1	6.1	6.7	9.9	15.1	19.9	23.4	27.6	28.8	25.2	19.6	13.8	8.7
	DBStd	8.21	2.13	2.73	3.10	3.15	2.18	2.13	1.99	1.61	2.55	2.74	2.94	2.56
	HDD10.0	323	122	99	41	2	0	0	0	0	0	2	57	
	HDD18.3	1536	378	326	262	104	10	0	0	0	18	139	299	
	CDD10.0	2923	2	7	38	155	307	403	546	582	456	296	116	16
	CDD18.3	1094	0	0	0	7	59	153	287	323	207	56	2	0
	CDH23.3	9919	0	0	0	14	194	832	3058	3886	1719	212	4	0
	CDH26.7	3442	0	0	0	1	15	139	1103	1662	502	21	0	0
Wind	WSAvg	3.8	4.2	4.0	4.1	4.0	3.7	3.4	3.8	3.7	3.5	3.5	3.7	4.1
Precipitation	PrecAvg	1447	48	68	99	103	149	192	155	112	201	147	107	64
	PrecMax	1935	122	167	182	226	446	370	406	544	445	354	350	166
	PrecMin	718	5	21	50	37	36	67	45	7	33	34	15	5
	PrecStd	298	30	37	34	45	95	78	94	112	111	84	70	38
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.4	18.5	20.8	25.2	28.4	31.5	35.0	35.6	32.9	28.6	23.4	18.8
		MCWB	12.0	14.8	14.8	16.8	18.5	23.5	25.2	25.6	24.1	22.7	17.6	14.6
	2%	DB	13.0	15.7	19.0	23.1	26.6	29.4	33.2	34.1	31.6	26.9	21.7	16.1
		MCWB	9.1	11.9	14.0	16.5	18.2	22.6	24.8	25.1	24.0	21.0	16.7	11.9
	5%	DB	11.5	13.5	17.3	21.7	25.4	28.2	32.1	33.0	30.6	25.6	20.3	14.5
		MCWB	7.4	9.3	12.6	15.7	18.0	21.8	25.0	25.0	23.6	19.8	15.6	10.4
	10%	DB	10.2	11.6	15.9	20.5	24.3	27.1	31.1	32.1	29.5	24.4	19.0	13.3
		MCWB	6.3	7.5	11.2	14.9	17.8	21.6	24.8	25.1	23.2	18.7	14.7	9.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.7	15.3	17.1	20.3	22.6	25.1	27.0	27.5	26.3	24.2	20.0	16.0
		MCDB	14.7	17.6	19.0	22.6	25.5	28.6	31.2	31.7	30.4	27.2	22.0	17.9
	2%	WB	9.9	12.5	14.9	18.6	21.4	24.4	26.3	26.8	25.5	22.5	18.0	12.8
		MCDB	11.9	15.3	17.8	21.1	24.0	27.7	30.7	31.3	29.5	25.8	20.3	15.3
	5%	WB	8.0	10.0	13.3	17.0	20.1	23.7	25.8	26.3	24.8	21.1	16.6	11.0
		MCDB	10.8	12.5	16.7	20.5	23.1	27.0	30.3	30.8	28.8	24.3	19.4	13.7
	10%	WB	6.6	8.1	11.7	15.8	19.1	23.0	25.4	25.9	24.2	19.7	15.2	9.7
		MCDB	9.8	11.1	15.4	19.5	22.8	26.1	29.7	30.3	28.2	23.2	18.4	12.8
Mean Daily Temperature Range	5% DB	MDBR	6.3	6.9	7.9	8.3	7.9	6.4	6.1	6.7	6.8	7.2	7.3	6.7
		MCDBR	8.8	9.7	10.0	9.5	9.4	7.7	7.3	7.8	7.7	8.3	8.6	8.9
		MCWBR	6.3	7.3	7.4	5.6	4.7	3.4	2.8	3.0	3.3	4.4	5.8	6.2
	5% WB	MCDBR	7.9	8.8	9.1	7.5	7.0	5.7	6.4	6.8	6.7	6.8	7.3	7.9
		MCWBR	6.8	7.8	8.1	5.8	5.2	3.1	2.8	3.0	3.5	4.7	6.2	6.7
Clear-Sky Solar Irradiance	taub	0.393	0.437	0.496	0.529	0.534	0.550	0.553	0.523	0.485	0.447	0.415	0.388	
	taud	2.203	2.092	1.945	1.915	1.976	2.013	2.052	2.131	2.208	2.230	2.226	2.230	
	Ebn at Noon	790	793	776	772	773	758	753	769	780	776	761	769	
	Edh at Noon	114	142	177	191	182	175	168	153	135	122	111	105	
All-Sky Solar Radiation	RadAvg	2.45	3.07	4.05	4.83	5.22	4.64	5.11	5.25	4.11	3.32	2.58	2.21	
	RadStd	0.18	0.23	0.30	0.45	0.53	0.46	0.62	0.45	0.35	0.34	0.22	0.14	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page